

Day 5

#4. Detecting Duplicate Records

#find duplicates

```
duplicates=df.duplicated()
```

```
print(duplicates)
```

#count duplicates:

```
print(df.duplicated().sum())
```

0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False
9	False
10	False
11	False
12	False
13	False
14	False
15	False
16	False
17	False
18	False
19	False
20	False
21	False
22	False
23	False
24	False
25	False
26	False
27	False
28	False
29	False
30	False
31	False
32	False
33	False
34	False
35	False
36	False
37	False
38	False
39	False

```
40    False
41    False
42    False
43    False
44    False
45    False
46    False
47    False
48    False
49    False
50    False
51    False
dtype: bool
0
```

#5. Dropping Duplicate Records

#Before

```
print(df.shape)
```

#Remove Duplicates

```
df.drop_duplicates(inplace=True)
```

#After

```
print(df.shape)
```

```
(52, 6)
```

```
(52, 6)
```

#6. String operations

#string methods work only on text columns.

#Remove Extra spaces

```
df["Student_Name"]=df["Student_Name"].str.strip()
```

#convert to uppercase

```
df["Student_Name"]=df["Student_Name"].str.upper()
```

```
print(df["Student_Name"].head())
```

#convert to title case

```
df["Student_Name"]=df["Student_Name"].str.title()
```

```
print(df["Student_Name"].head())
```

```
0    STUDENT1
```

```
1    STUDENT2
```

```
2    STUDENT3
```

```
3    STUDENT4
```

```
4    STUDENT5
```

```
Name: Student_Name, dtype: object
```

```
0    Student1
```

```
1    Student2
```

```
2    Student3
```

```
3    Student4
```

```
4    Student5
```

```
Name: Student_Name, dtype: object
```

```

#Replace Text
#Replace Department Name
df["Department"]=df["Department"].str.replace(
    "CSE",
    "Computer Science"
)
print(df["Department"].head())
# replace multiple values
df["Department"]=df["Department"].replace({
    "CSE":"Computer science",
    "ECE":"Electronics and communi eng",
    "EEE":"electrical and electronic",
    "IT":"information tech",
    "MECH":"mechanical"
})
print(df["Department"].head())

```

0	IT
1	ECE
2	EEE
3	MECH
4	Computer Science

Name: Department, dtype: object

0	information tech
1	Electronics and communi eng
2	electrical and electronic
3	mechanical
4	Computer Science

Name: Department, dtype: object

```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in c:\users\devad\anaconda3\lib\site-packages (3.10.0)
```

```
Requirement already satisfied: contourpy>=1.0.1 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib) (1.3.1)
```

```
Requirement already satisfied: cyclor>=0.10 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
```

```
Requirement already satisfied: fonttools>=4.22.0 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib) (4.55.3)
```

```
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib) (1.4.8)
```

```
Requirement already satisfied: numpy>=1.23 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib) (2.1.3)
```

```
Requirement already satisfied: packaging>=20.0 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib) (24.2)
```

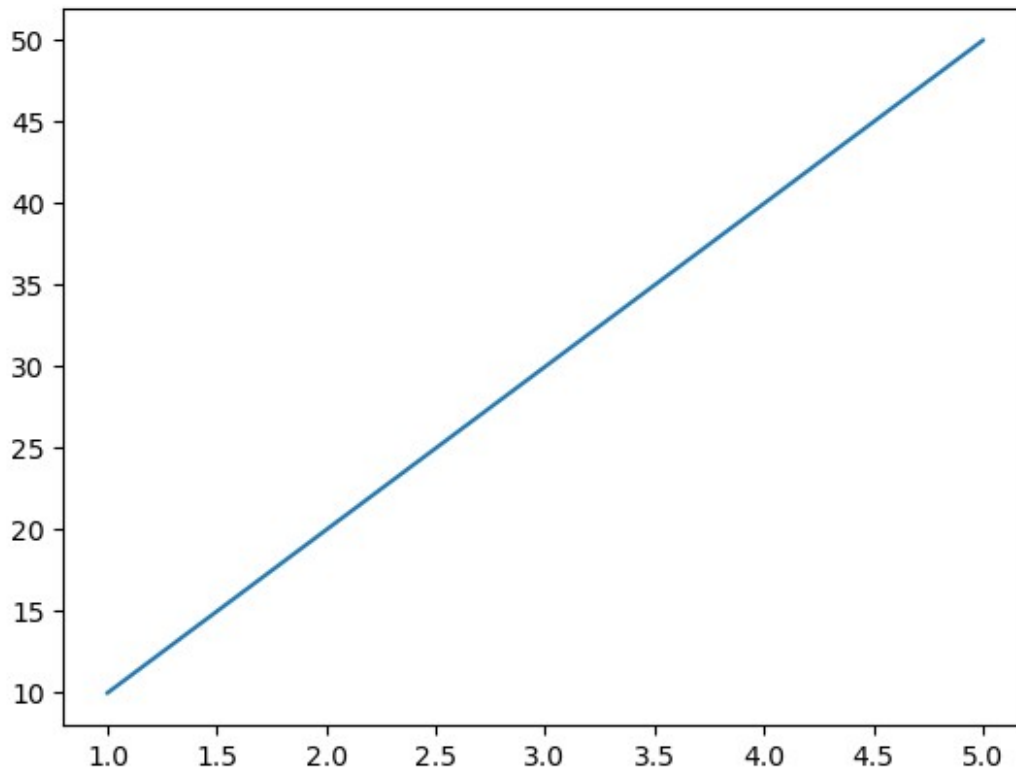
```
Requirement already satisfied: pillow>=8 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib) (11.1.0)
```

```
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib) (3.2.0)
```

```
Requirement already satisfied: python-dateutil>=2.7 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)
```

```
Requirement already satisfied: six>=1.5 in c:\users\devad\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,30,40,50]
plt.plot(x,y)
plt.show()
```



```
#figure and axes
#these are the most important concepts.
#Figure-The entire chart window
#Axes-the area where the graph is drawn
import matplotlib.pyplot as plt
fig,ax=plt.subplots()
ax.plot([1,2,3],[10,20,30])
#Adding a Title
plt.title("Monthly Sales")
#adding x and y labels
plt.xlabel("Month")
plt.ylabel("Sales")
#changing line color
plt.plot(x,y,color="green")
#changing line style
plt.plot(x,y,linestyle="-.")

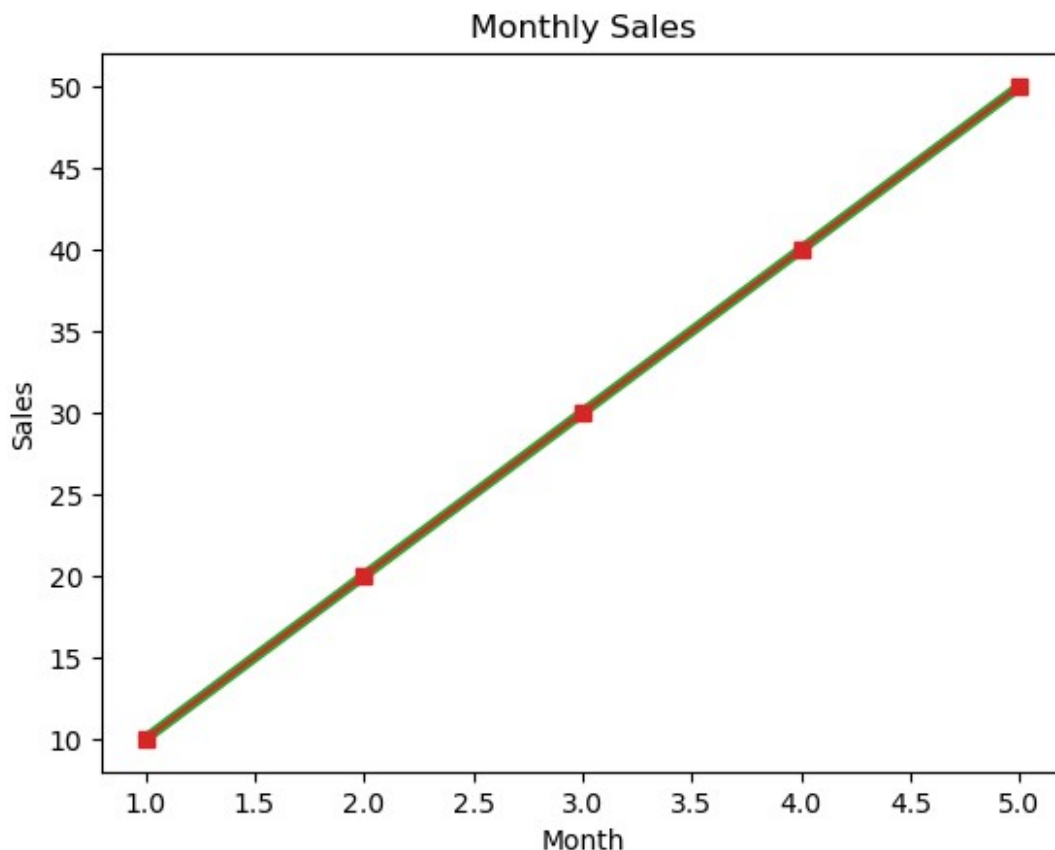
#styles:
# '-' Solid
# '--' Dashed
# ':' Dotted
# '-.' Dash Dot

#changing Line Width
plt.plot(x,y,linewidth=4)
```

```

#thicker line
#adding markers
plt.plot(x,y,marker="s")
#Output
# o----o----o----o
#Marker types:
# 'o' Circle
# 's' Square
# '^' Triangle
# '*' Star
plt.show()

```



```
!pip install seaborn
```

```

Requirement already satisfied: seaborn in c:\users\devad\anaconda3\
lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\devad\
anaconda3\lib\site-packages (from seaborn) (2.1.3)
Requirement already satisfied: pandas>=1.2 in c:\users\devad\
anaconda3\lib\site-packages (from seaborn) (2.2.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\
devad\anaconda3\lib\site-packages (from seaborn) (3.10.0)

```

Requirement already satisfied: contourpy>=1.0.1 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.1)

Requirement already satisfied: cycler>=0.10 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.55.3)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.8)

Requirement already satisfied: packaging>=20.0 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (24.2)

Requirement already satisfied: pillow>=8 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (11.1.0)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.0)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\devad\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)

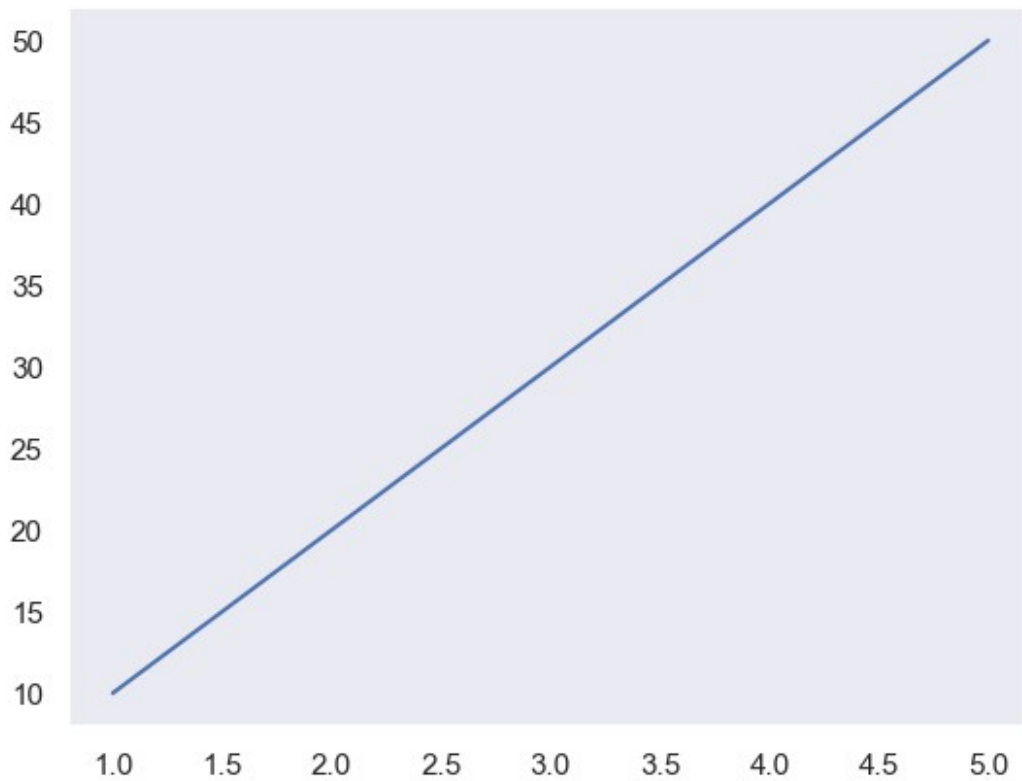
Requirement already satisfied: pytz>=2020.1 in c:\users\devad\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2024.1)

Requirement already satisfied: tzdata>=2022.7 in c:\users\devad\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)

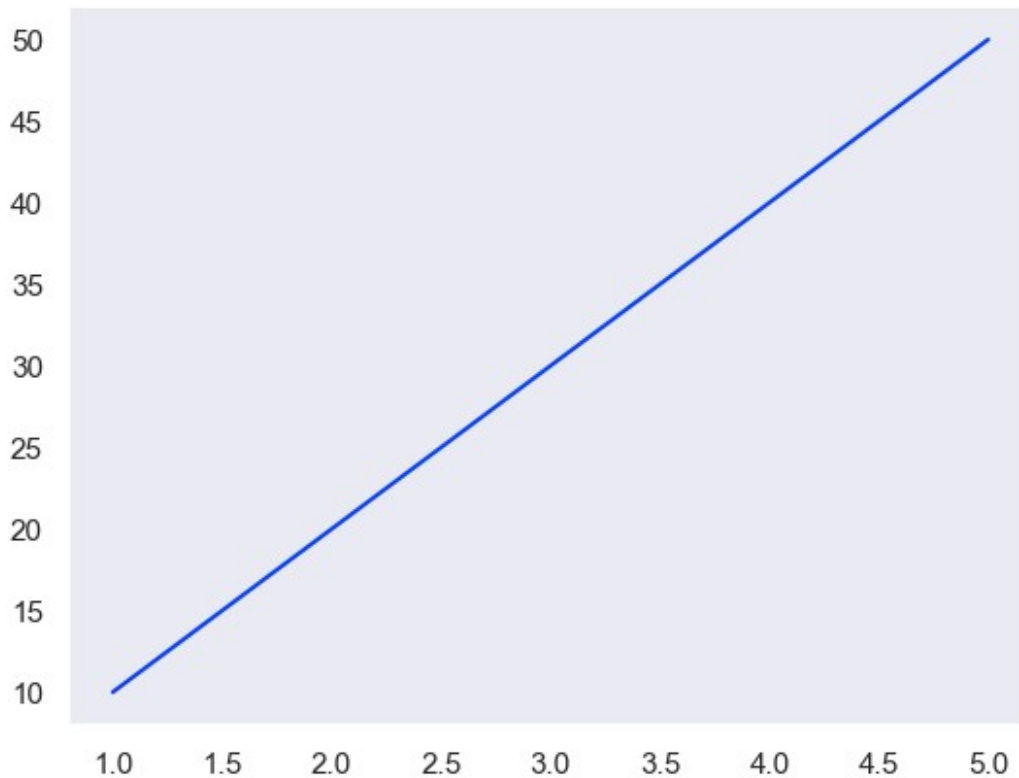
Requirement already satisfied: six>=1.5 in c:\users\devad\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)

```
import seaborn as sns
import matplotlib.pyplot as plt

x=[1,2,3,4,5]
y=[10,20,30,40,50]
#themes in seaborn
#theme control the appearance of charts
#default Theme
sns.set_theme()
#Dark grid theme
sns.set_style("dark")
#available styles
#darkgrid
#whitegrid
#dark
#white
#tick
sns.lineplot(x=x,y=y)
plt.show()
```



```
#color palette
#A palette is a collection of colors
#view current pallete
sns.set_palette("bright")
sns.lineplot(
    x=[1,2,3,4,5],
    y=[10,20,30,40,50]
)
plt.show()
#available sytle
#deep
#muted
#pastel
#bright
#dark
#colorblind
```



```
#Built-in Dataset Support
#One of Seaborn's biggest advantages is built in datasets.
#view available Datasets
import seaborn as sns
print(sns.get_dataset_names())

['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes',
 'diamonds', 'dots', 'dowjones', 'exercise', 'flights', 'fmri',
 'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets',
 'seance', 'taxi', 'tips', 'titanic']
```

```
#load Titanic Dataset
import seaborn as sns
df=sns.load_dataset("titanic")
print(df.head)
```

	<bound method NDFrame.head of	survived	pclass	sex	age
	sibsp parch fare embarked class \				
0	0 3 male 22.0	1	0	7.2500	S
Third					
1	1 1 female 38.0	1	0	71.2833	C
First					
2	1 3 female 26.0	0	0	7.9250	S
Third					
3	1 1 female 35.0	1	0	53.1000	S
First					

4	0	3	male	35.0	0	0	8.0500	S
Third								
...	...	...	...	...	...	...	...	...
...								
886	0	2	male	27.0	0	0	13.0000	S
Second								
887	1	1	female	19.0	0	0	30.0000	S
First								
888	0	3	female	NaN	1	2	23.4500	S
Third								
889	1	1	male	26.0	0	0	30.0000	C
First								
890	0	3	male	32.0	0	0	7.7500	Q
Third								

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True
...	...	...	...	...	...	...
886	man	True	NaN	Southampton	no	True
887	woman	False	B	Southampton	yes	True
888	woman	False	NaN	Southampton	no	False
889	man	True	C	Cherbourg	yes	True
890	man	True	NaN	Queenstown	no	True

[891 rows x 15 columns]>

#Linechart

#what is a line chart?

#a line chart connects data points

```
import matplotlib.pyplot as plt
```

```
months=["Jan","Feb","mar","Apr"]
```

```
sales=[100,150,200,250]
```

```
plt.plot(months,sales,marker="o")
```

```
plt.title("Monthly Sales")
```

```
plt.xlabel("Month")
```

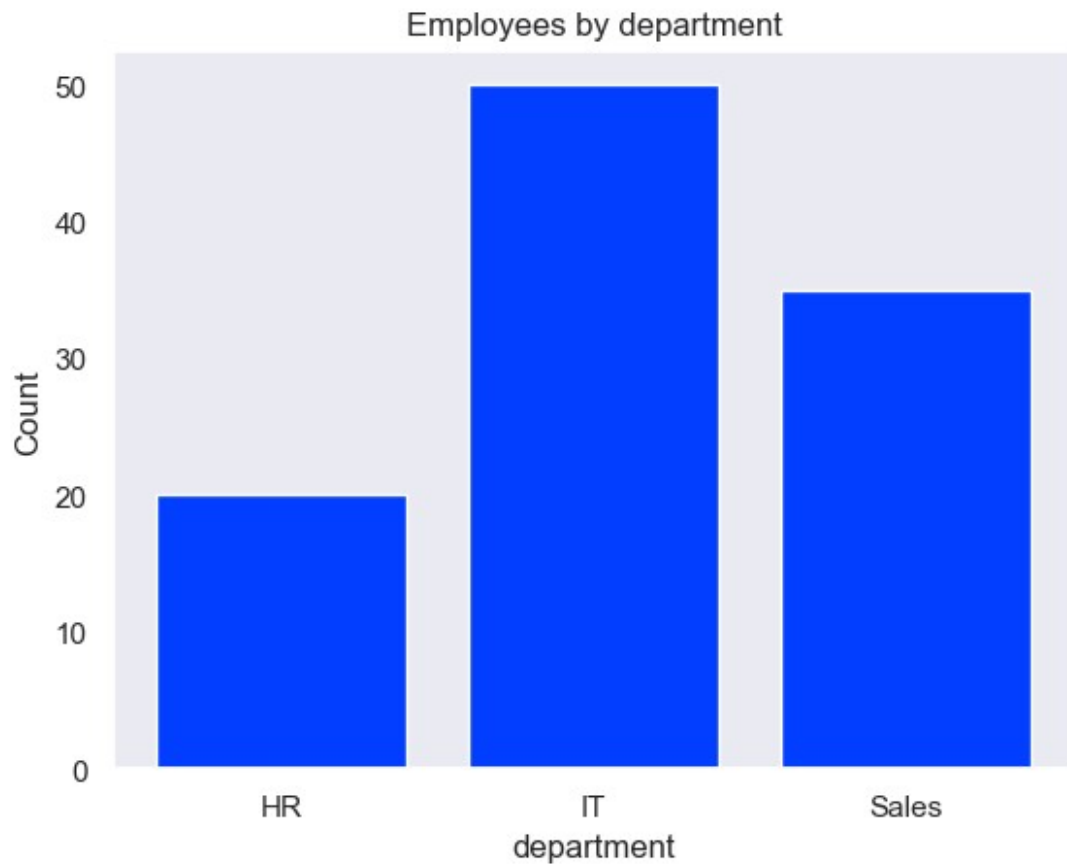
```
plt.ylabel("Sales")
```

```
plt.show
```

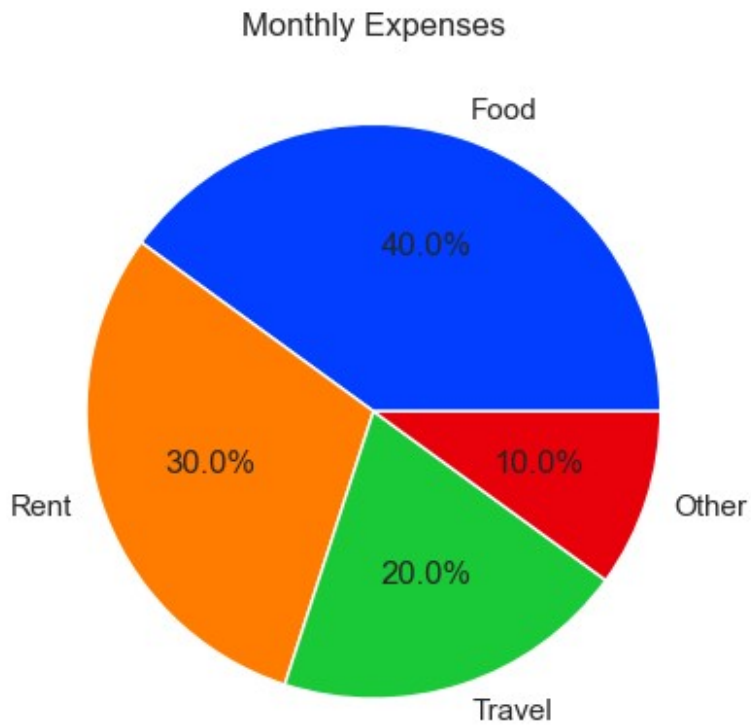
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



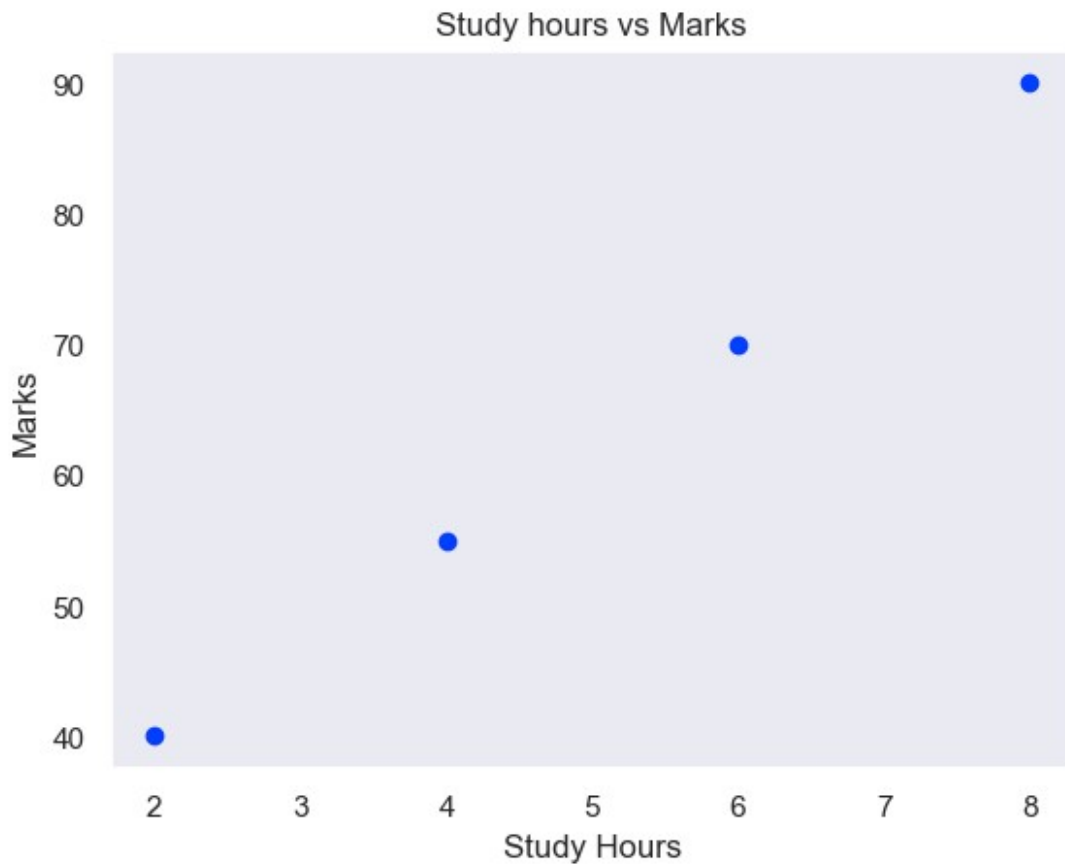
```
#bar chart
import matplotlib.pyplot as plt
dept=["HR","IT","Sales"]
employees=[20,50,35]
plt.bar(dept,employees)
plt.title("Employees by department")
plt.xlabel("department")
plt.ylabel("Count")
plt.show()
```



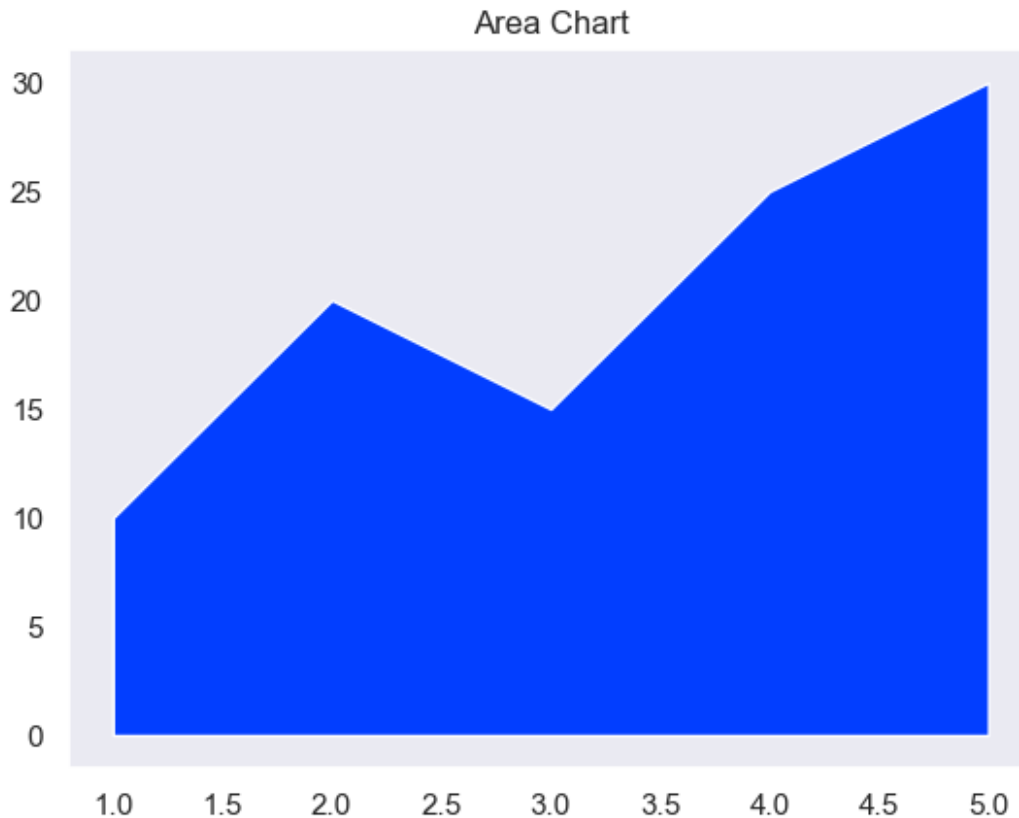
```
#pie chart
import matplotlib.pyplot as plt
labels=["Food","Rent","Travel","Other"]
values=[40,30,20,10]
plt.pie(
    values,
    labels=labels,
    autopct="%1.1f%%"
)
plt.title("Monthly Expenses")
plt.show()
```



```
#scatter plot
import matplotlib.pyplot as plt
hours=[2,4,6,8]
marks=[40,55,70,90]
plt.scatter(hours,marks)
plt.title("Study hours vs Marks")
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.show()
```

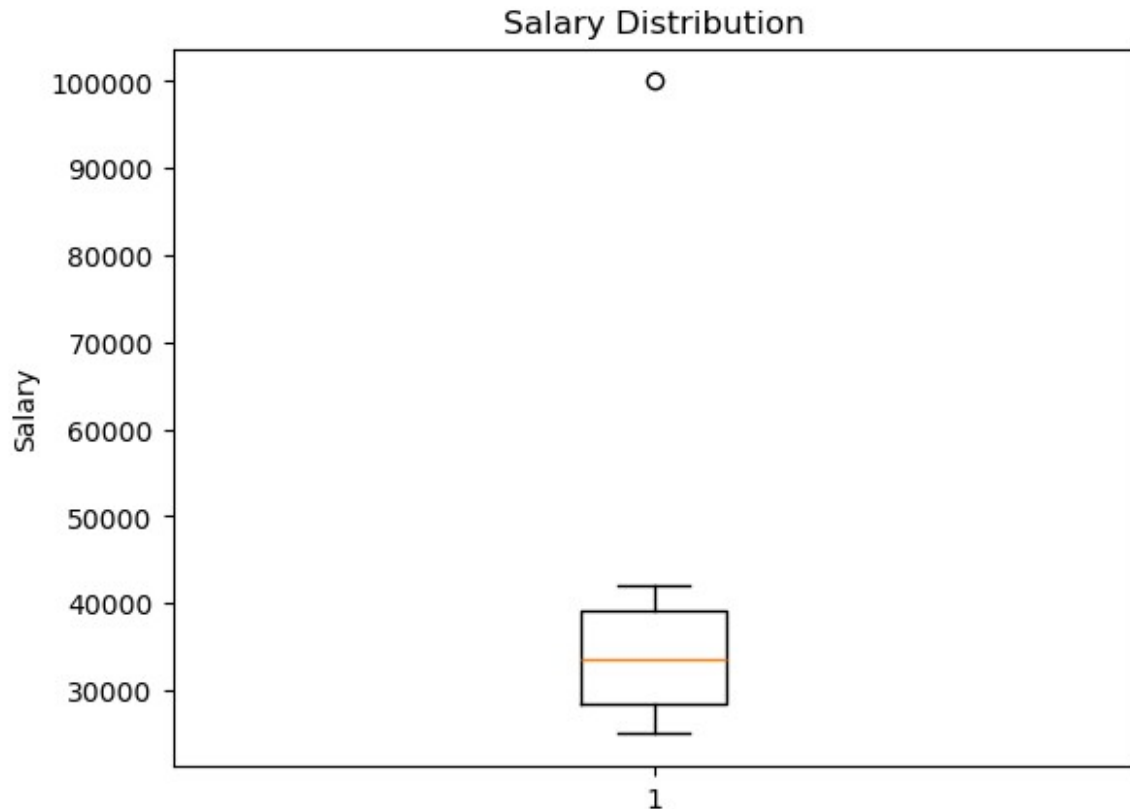


```
#Area Chart
#purpose: shows cumulative totals over time
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,15,25,30]
plt.fill_between(x,y)
plt.title("Area Chart")
plt.show()
```



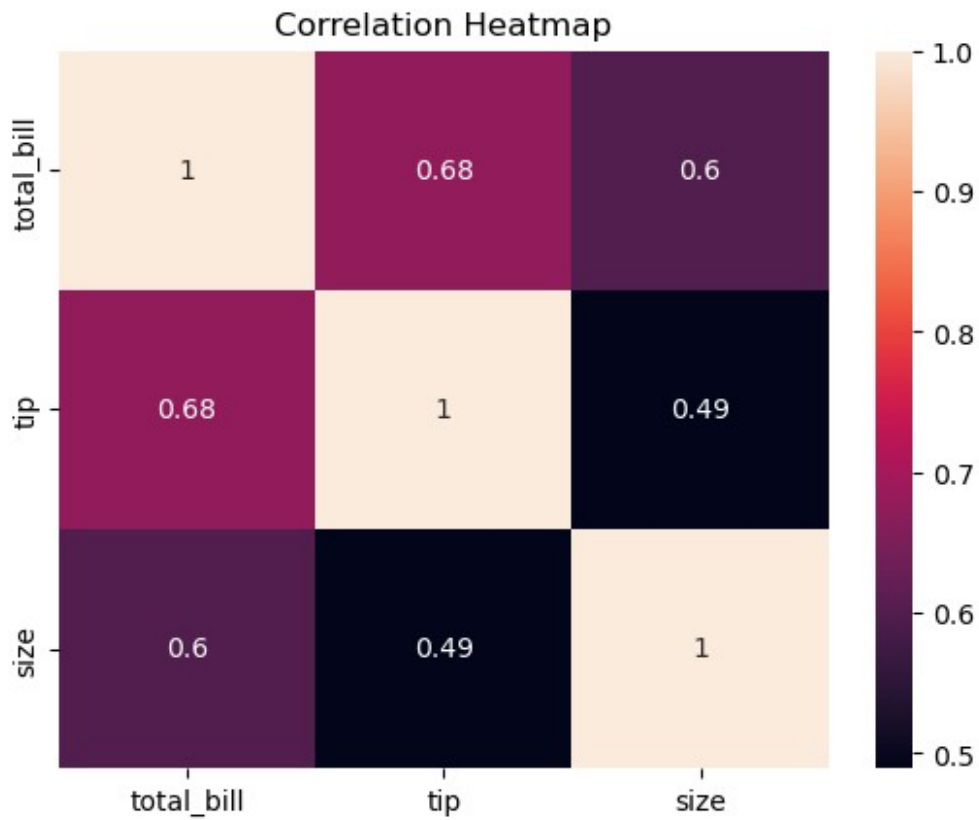
```
#Box plot
#A Box Plot is used to visualize the distribution of numerical data
and identify outliers
#components of a box plot
#Min-----Q1-----[Median]-----Q3-----Max
#
# Lower Whisker Upper Whisker
#minimum:Smallest value(excluding outliers)
#Q1 (25%) : First quartile
#medium (50%): middle value
#Q3 (75%): Third quartile
#Maximum: Largest value(excluding outliers)
#Outliers: values far from the rest of the data,shown as individual
points

import matplotlib.pyplot as plt
salary=[25000,28000,30000,32000,35000,37000,40000,42000,25000,100000]
plt.boxplot(salary)
plt.title("Salary Distribution")
plt.ylabel("Salary")
plt.show()
```

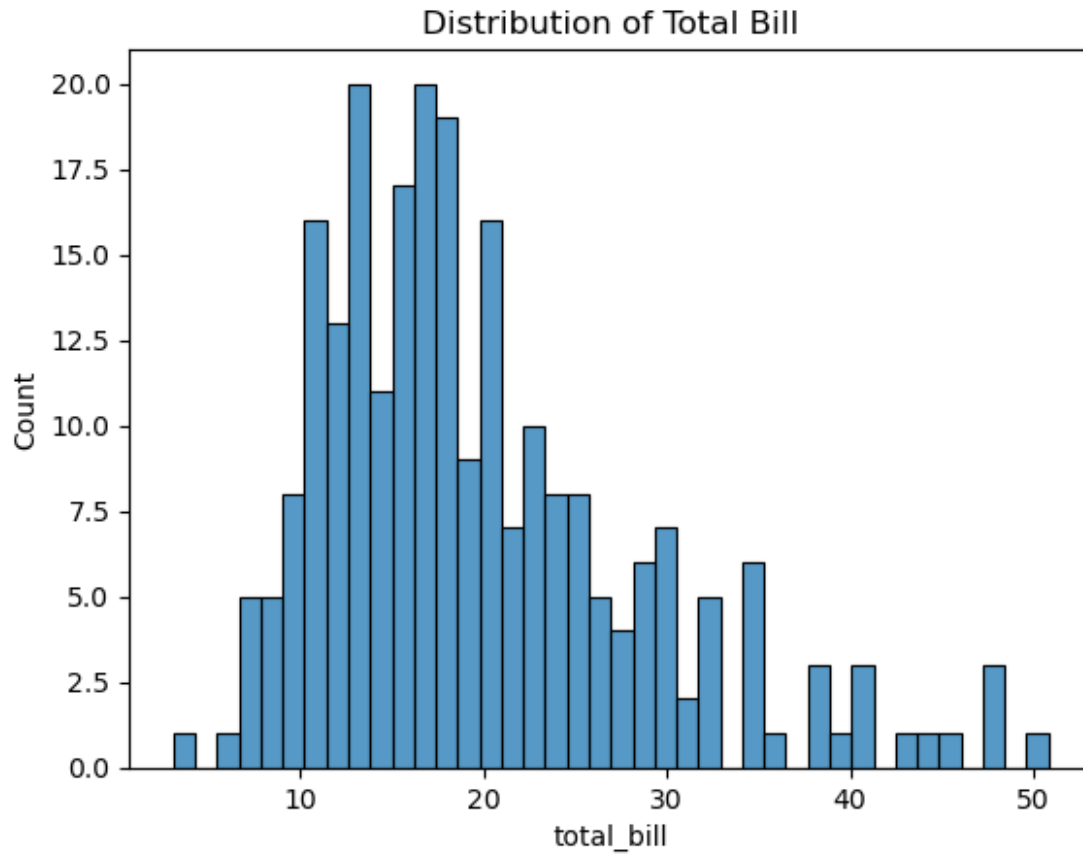


```
#Heatmap
#A heatmap uses colors to represent values in a table or matrix
#In data analysis, heatmaps are commonly used to visualize correlations
between numerical columns
#correlation values
#Value meaning
#+1    ---->Strong Positive Relationship
#0     ---->No relationship
#-1    ---->Strong Negative Relationship

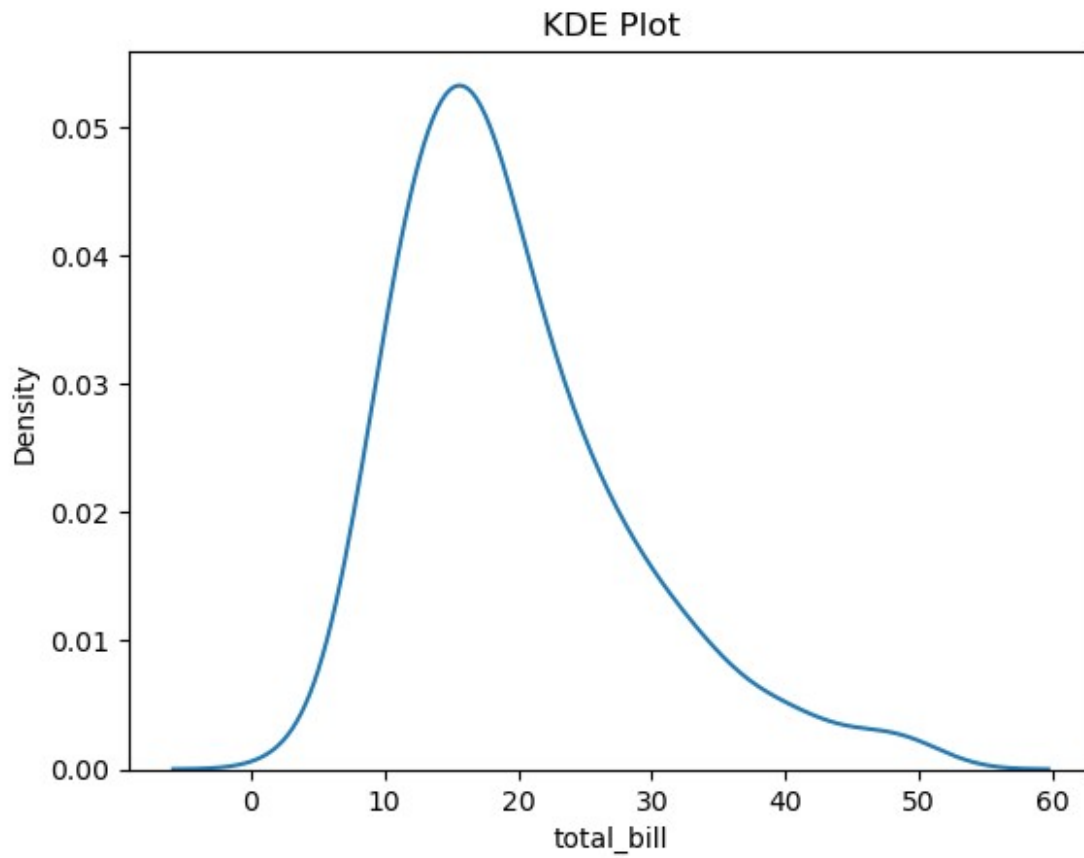
#correlation heatmap with seaborn
import seaborn as sns
import matplotlib.pyplot as plt
#Load dataset
tips=sns.load_dataset("tips")
#correlation matrix
corr=tips.corr(numeric_only=True)
#Heatmap
sns.heatmap(corr,annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



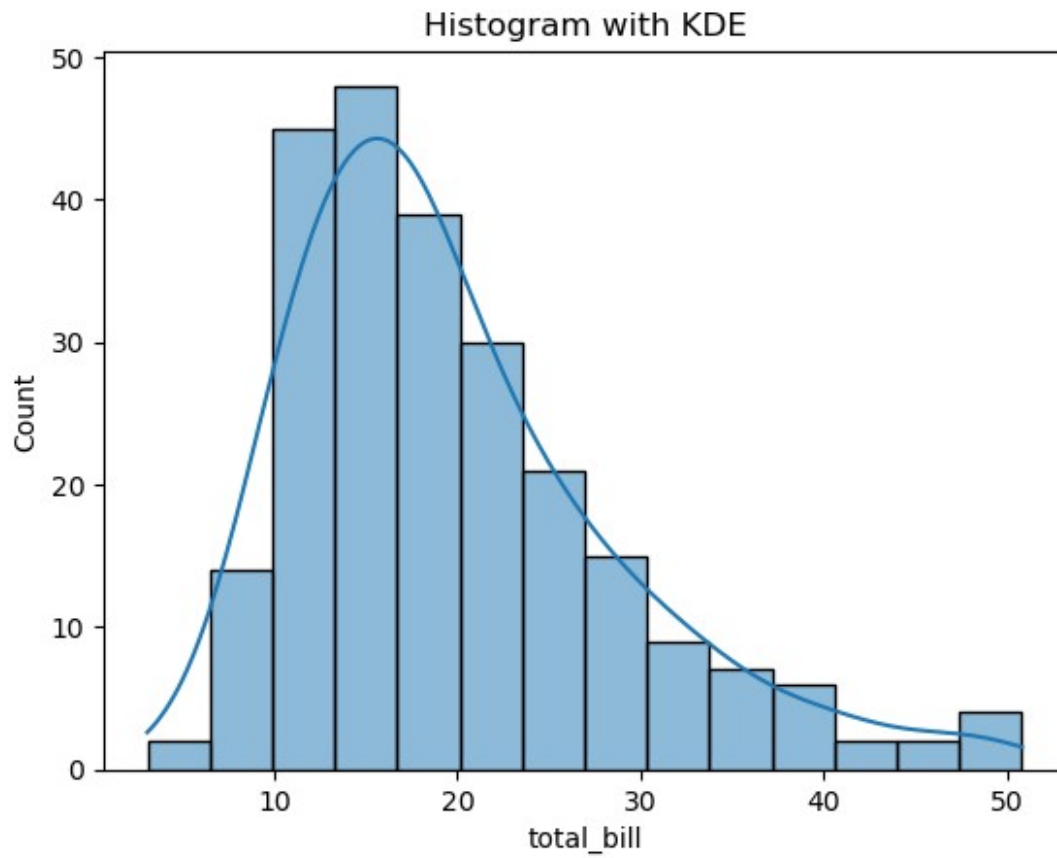
```
#histogram using seaborn
sns.histplot(
    data=tips,
    x="total_bill",
    bins=40
)
plt.title("Distribution of Total Bill")
plt.show()
#sns.histplot()
#creates histogram
```



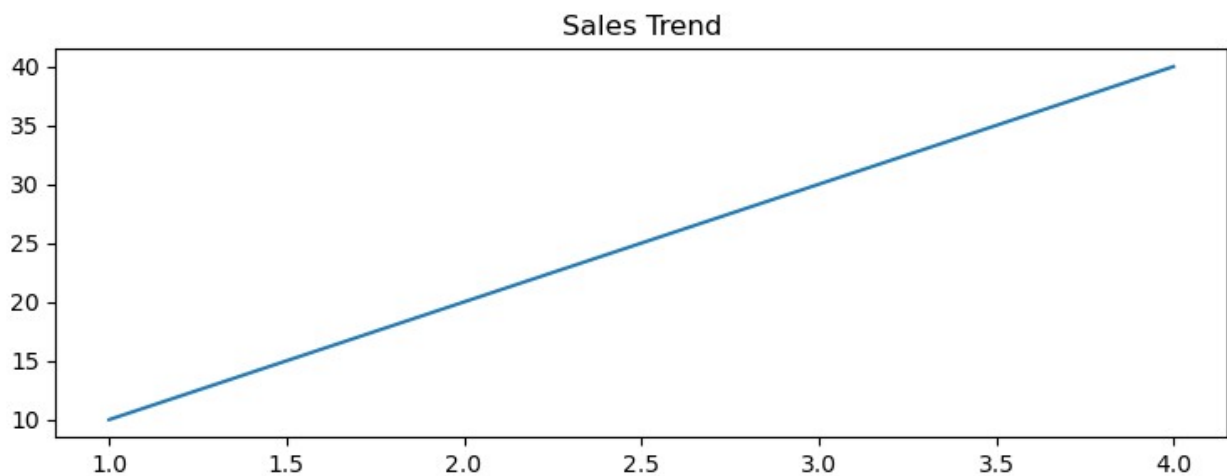
```
#KDE Plot (Kernel Density Estimation)  
#A KDE Plot is a smooth version of a histogram  
#Instead of bars,it shows a smooth curve representing data  
distribution  
#When to use KDE? understanding data distribution identifying peaks  
comparing distributions statistical analysis  
  
sns.kdeplot(  
    data=tips,x="total_bill"  
)  
plt.title("KDE Plot")  
plt.show()
```



```
sns.histplot(  
    data=tips,  
    x="total_bill",  
    kde=True  
)  
plt.title("Histogram with KDE")  
plt.show()  
  
#kde=True  
#adds a smooth density curve on top of the histogram
```



```
#figure size cintrols the width and height of the chart  
import matplotlib.pyplot as plt  
plt.figure(figsize=(9,3))  
plt.plot([1,2,3,4],[10,20,30,40])  
plt.title("Sales Trend")  
plt.show()
```



```
#subplots allow multiple charts in a single figure.Syntax:plt.subplot(rows,columns,position)  
#2 charts
```

```
#2 charts
```

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(8,4))
```

```
# First Plot
```

```
plt.subplot(1,2,2)
```

```
plt.plot([1,2,3], [10,20,30])
```

```
plt.title("Line Chart")
```

```
# Second Plot
```

```
plt.subplot(1,2,1)
```

```
plt.bar(['A','B','C'], [5,8,6])
```

```
plt.title("Bar Chart")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
#Example: Four Charts
```

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(10,8))
```

```
plt.subplot(2,2,1)
```

```
plt.plot([1,2,3], [10,20,30])
```

```
plt.title("Line")
```

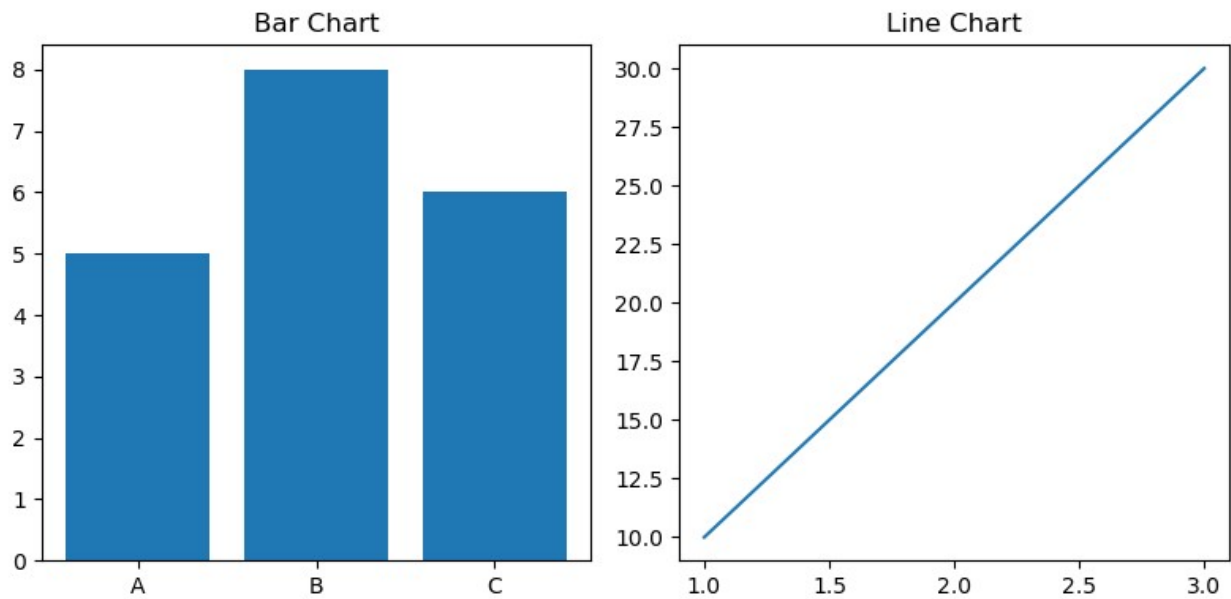
```
plt.subplot(2,2,2)
```

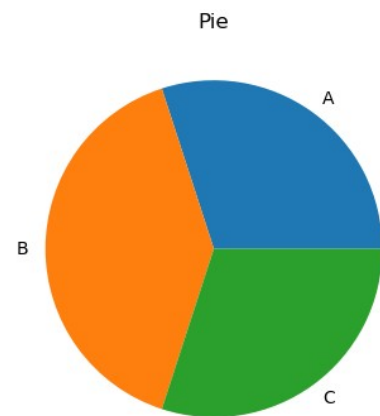
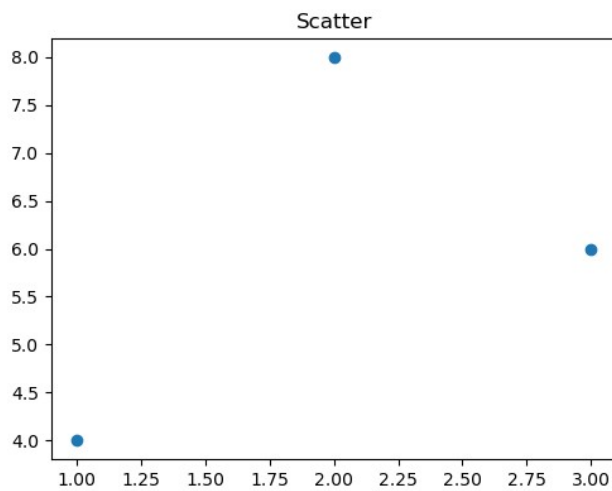
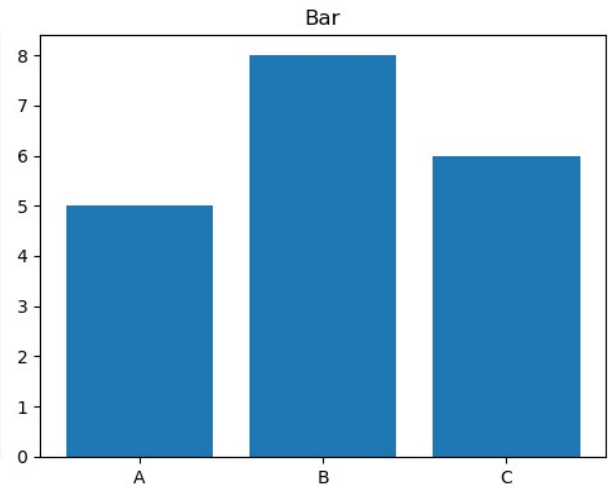
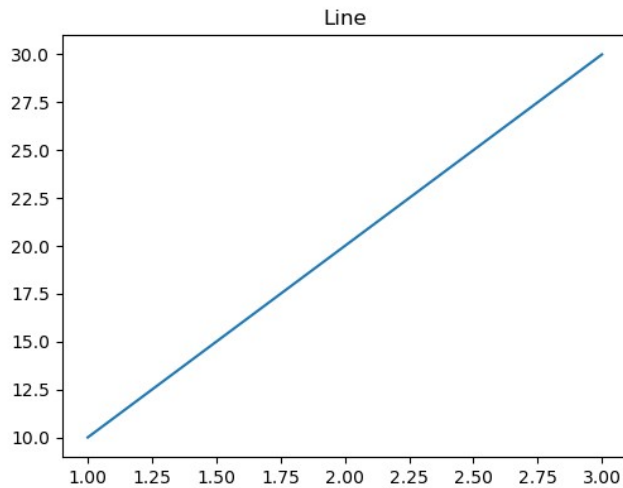
```
plt.bar(['A','B','C'], [5,8,6])
```

```
plt.title("Bar")
```

```
plt.subplot(2,2,3)
```

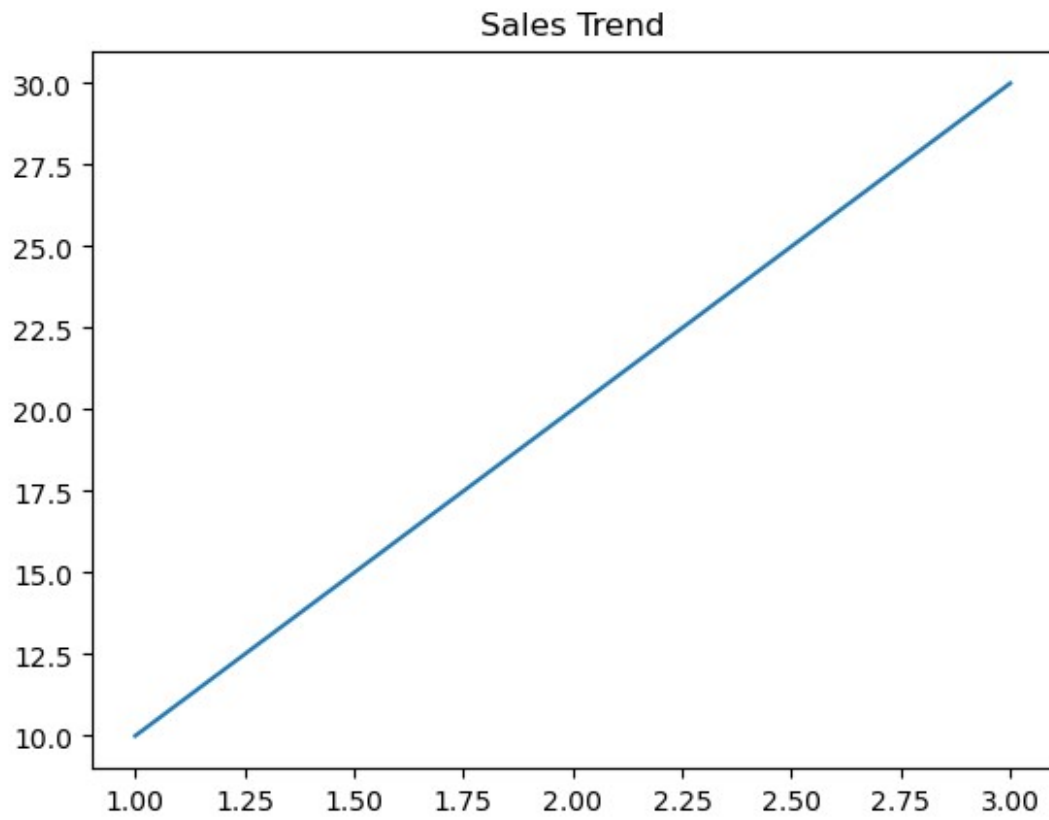
```
plt.scatter([1,2,3], [4,8,6])
plt.title("Scatter")
plt.subplot(2,2,4)
plt.pie([30,40,30], labels=['A','B','C'])
plt.title("Pie")
plt.tight_layout()
plt.show()
```





```
import matplotlib.pyplot as plt
plt.plot([1,2,3],[10,20,30])
plt.title("Sales Trend")
plt.savefig("Sales_chart.png")
plt.show()

#save with high quality
plt.savefig(
    "chart.png",dpi=300
)
```



<Figure size 640x480 with 0 Axes>